Method Overloading

Exercise 1: Add Numbers

Create a class Calculator and overload a method named add() to:

- Add two int numbers
- Add three int numbers
- Add two double numbers

Example:

add(10, 20);

add(10, 20, 30);

add(5.5, 4.2);

Exercise 2: Find Area

Create a class AreaCalculator and overload a method named area() to calculate:

- Area of a square: area(int side)
- Area of a rectangle: area(int length, int width)
- Area of a circle: area(double radius)

Expected formulas:

Square = side × side

Rectangle = length × width

Circle = π × radius × radius

Exercise 3: Display Information

Create a class Student and overload a method called display().

It should support:

display(String name);

display(String name, int age);

display(String name, int age, double cgpa);

Example output:

Name: Rahim

Name: Karim, Age: 20

Name: Sakib, Age: 22, CGPA: 3.75

Exercise 4: Find Maximum

Create overloaded methods called max() that find:

- Maximum between two integers
- Maximum among three integers
- Maximum between two doubles

Example:

max(10, 25);

max(10, 50, 30);

max(4.5, 8.7);

Exercise 5: Calculate Salary

Create a class Employee and overload a method named calculateSalary().

Use:

calculateSalary(double basicSalary);

calculateSalary(double basicSalary, double bonus);

calculateSalary(double basicSalary, double bonus, double overtime);

The methods should return the employee's total salary.

Exercise 6: Print Different Data Types

Create overloaded methods named print() that accept:

print(int number);

print(double number);

print(String text);

print(boolean value);

Example:

print(100);

print(45.67);

print("Java");

print(true);

Exercise 7: Volume Calculator

Create a class VolumeCalculator and overload volume() to calculate the volume of:

- Cube
- Rectangular box
- Cylinder

Suggested methods:

volume(int side);

volume(int length, int width, int height);

volume(double radius, double height);

Exercise 8: Simple Bank Deposit

Create a class BankAccount with overloaded deposit() methods:

deposit(double amount);

deposit(double amount, String description);

deposit(double amount, String description, int transactionId);

Print appropriate transaction information from each method.

Exercise 9: Convert Measurements

Create overloaded methods named convert().

For example:

convert(int meter); // meter to centimeter

convert(double kilometer); // kilometer to meter

convert(double value, String unit);

For the third method, the user might call:

convert(5.0, "km");

convert(10.0, "meter");

Exercise 10: Method Overloading with Different Parameter Order

Create two overloaded methods:

show(String name, int age);

show(int age, String name);

Call:

show("John", 25);

show(25, "John");

Exercise 11: Shopping Bill

Create a class ShoppingCart and overload calculateBill().

calculateBill(double price);

calculateBill(double price, int quantity);

calculateBill(double price, int quantity, double discount);